

Andres Perez

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EDUCATION

University of Notre Dame

Bachelor of Science in Computer Engineering | Minor: Engineering Corporate Practice

Graduation Date: May 2028

GPA: 3.7/4.0

Engineering Study Abroad: London, England

May 2025 – Jun. 2025

Notre Dame Aerospace Career Trek: Northrop Gruman, JPL, SpaceX

Jul. 2025

Activities: IrishSat (CubeSat) Team, Career Assistant Intern

Enrolled Courses: Data Structures, Discrete Math, Intro. to ECE, Engineering Physics II, Linear Algebra & Differential Equations

RESEARCH EXPERIENCE

Department of Electrical Engineering

Aug. 2025 – Present

- Developing a real-time Python software interface for tracking and visualizing GPS-enabled test resources, for drones and software-defined radios, that parses NMEA strings to extract GPS data and displays it on a live map using the folium library, supporting various research experiments.
- Creating weekly PowerPoints to summarize project progress and update next goals.

Department of Computer Science and Engineering

Aug. 2024 – Present

- Researched under Professor Chaoli Wang by data labeling image data, downloading GSV data of households, and surveying heat dissipation in household.

ENGINEERING PROJECTS AND LEADERSHIP

Notre Dame, IN

IrishSat's Gravitational Orbital Attitude Thermal Lab

Electronics Lead

Aug. 2024 – Present

- Appointed as incoming Electronics Lead to oversee embedded systems development and PCB design initiatives for testing equipment, particularly a Helmholtz cage project, coordinating hardware-software integration efforts across the cubesat projects.
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, integrating PCB design (KiCad), control algorithms (MATLAB), and embedded Linux systems (Raspberry Pi) into a unified testing platform.
- Assembled a custom 2S2P battery pack using 18650 lithium-ion cells to make a 7.2V 7000 mAh battery pack. The completed pack was then integrated with a Battery Management System (BMS) to validate its nominal values.

Hardware-Accelerated Human Activity Classifier | Independent Project

Long Beach, CA

- Built a wireless data acquisition system using two Arduino Uno R3s, an MPU-9250 IMU sensor, and nRF24L01 RF transceivers.
- Developed a C++ sketch to sample raw accelerometer and gyroscope data, format it, and transmit it wirelessly to a receiving station. Along with a Python script to receive, parse, and log the real-time sensor data, generating a custom labeled dataset for walking and sitting.
- Trained a 1D Convolutional Neural Network in PyTorch to classify human activities (walking, sitting), achieving a peak validation accuracy of 96.7% on the test set.
- Visualized training and testing accuracy against learning rate, window size, and optimizer to demonstrate model stability and performance trends, documenting the findings.

Helmholtz Cage Driver PCB

Notre Dame, IN

- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino, magnetometers, and H-bridge circuits, reducing assembly complexity.
- Amended the generated Bill of Materials and Component Placement List files export to JLCPCB for manufacturing.

Air Quality Sensor Housing | Engineering Design

Notre Dame, IN

- Collaborated with a team of four engineering students to develop and iterate on sensor housing with a rack-and-pinion opening mechanism actuated by a servo motor and ESP32-based microcontroller.
- Designed and prototyped with SolidWorks in two design phases, optimizing for form, fit, function, and durability through iterative testing and feedback.

Portfolio Website | andres-perez.github.io

Long Beach, CA

- Developed a student and engineering project portfolio website with Astro and Tailwind CSS frameworks as the front end.

TECHNICAL AND LANGUAGE SKILLS

Tools: Microsoft Office Suite, KiCad, Altium, Vim, SolidWorks, Fusion 360

Skills: Technical Writing, Electronics Soldering

Programming: C, C++, Python, Git, MATLAB, Astro

Language: English (Native), Spanish (Bilingual)

Hobbies: Japanese(N4-N5), Chess (1000+ Rapid)